

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of
Science and Commerce
Department of Information Technology (Bsc.IT Semester
III)

Python for Data Science Practical

Practical - III

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PART – A: Theory

Python provides built-in modules to work with **date**, **time**, and **calendar**. These modules help programmers display the current date, convert timestamps into readable dates, generate calendars, and compare dates. They are widely used in attendance systems, scheduling applications, event management, and report generation.

1. Working with Date and Time (datetime Module)

The datetime module provides classes to work with dates and time. It allows us to get the current date and time, format them, and perform calculations.

Syntax:

```
from datetime import datetime  
today = datetime.now()
```

Explanation:

- from datetime import datetime imports the datetime class.
- datetime.now() returns the current system date and time.

2. Working with Timestamps (time Module)

A timestamp is the number of seconds elapsed since **January 1, 1970 (Unix Epoch)**. The time module converts timestamps into a readable date and time.

Syntax:

```
import time
time.ctime(timestamp)
```

Explanation:

- ctime() converts a timestamp into a human-readable date and time.

3. Displaying a Calendar (calendar Module)

The calendar module is used to display monthly and yearly calendars in Python.

Syntax:

```
import calendar
calendar.month(year, month)
```

Explanation:

- month() displays the calendar for a given month and year.
- calendar() displays the complete calendar for a year.

4. Comparing Two Dates

Python allows comparison of dates using comparison operators or by calculating the difference between dates.

Syntax:

```
date1 > date2
date2 - date1
```

Explanation:

- > checks which date is later.
- - returns the difference between two dates.

PART – B Questions:

a. Write a python code to print current date in different format.

Code :

```

from datetime import datetime
d=datetime.now()
print(d.strftime("%d-%m-%Y"))
print(d.strftime("%d-%m-%y"))
print(d.strftime("%B-%m-%Y"))
print("HARSHAL____S036")

```

Output :

```

-----
29-07-2026
29-07-26
July-07-2026
HARSHAL____S036

```

b. Write a python code to convert time stamp to date stamp.

Code :

```

import time
t=time.time()
print(t)
print(time.ctime(t))
print("HARSHAL____S036")

```

Output :

```

-----
1785313651.7904575
Wed Jul 29 13:57:31 2026
HARSHAL____S036

```

c. Write a python code to develop calendar module.

Code :

```

import calendar
print(calendar.month(2026,7))
print("HARSHAL____S036")

```

Output :

```

-----
      July 2026
Mo Tu We Th Fr Sa Su
      1  2  3  4  5
  6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30 31

```

```

HARSHAL____S036

```

d. Write a python code to compare two dates.

Code :

```

-----
from datetime import date
d1=date(2026,7,29)
d2=date(2026,9,14)
print(d2>d1)
print(d2-d1)
print("HARSHAL____S036")

```

Output :

```

-----
True
47 days, 0:00:00
HARSHAL____S036
>

```

